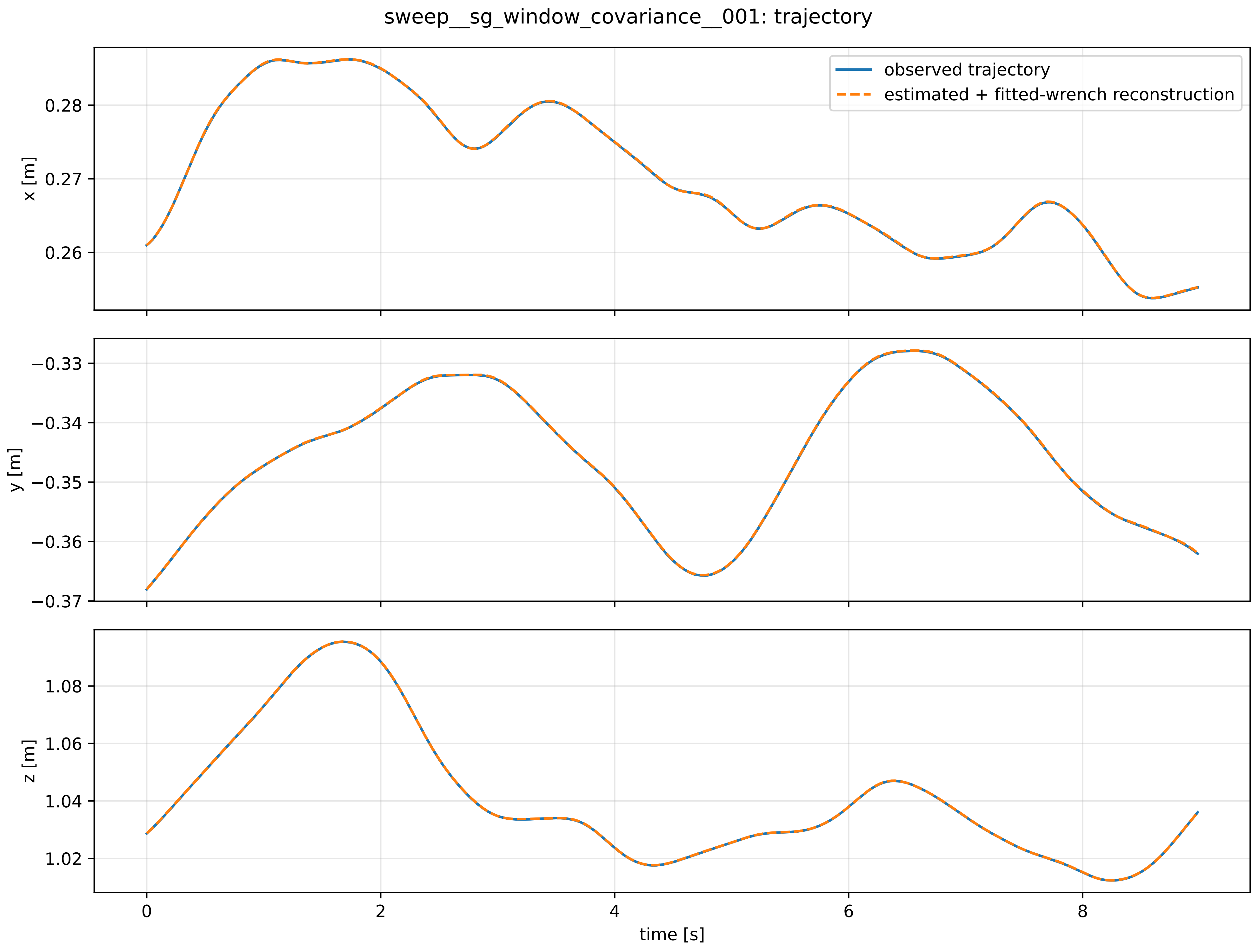
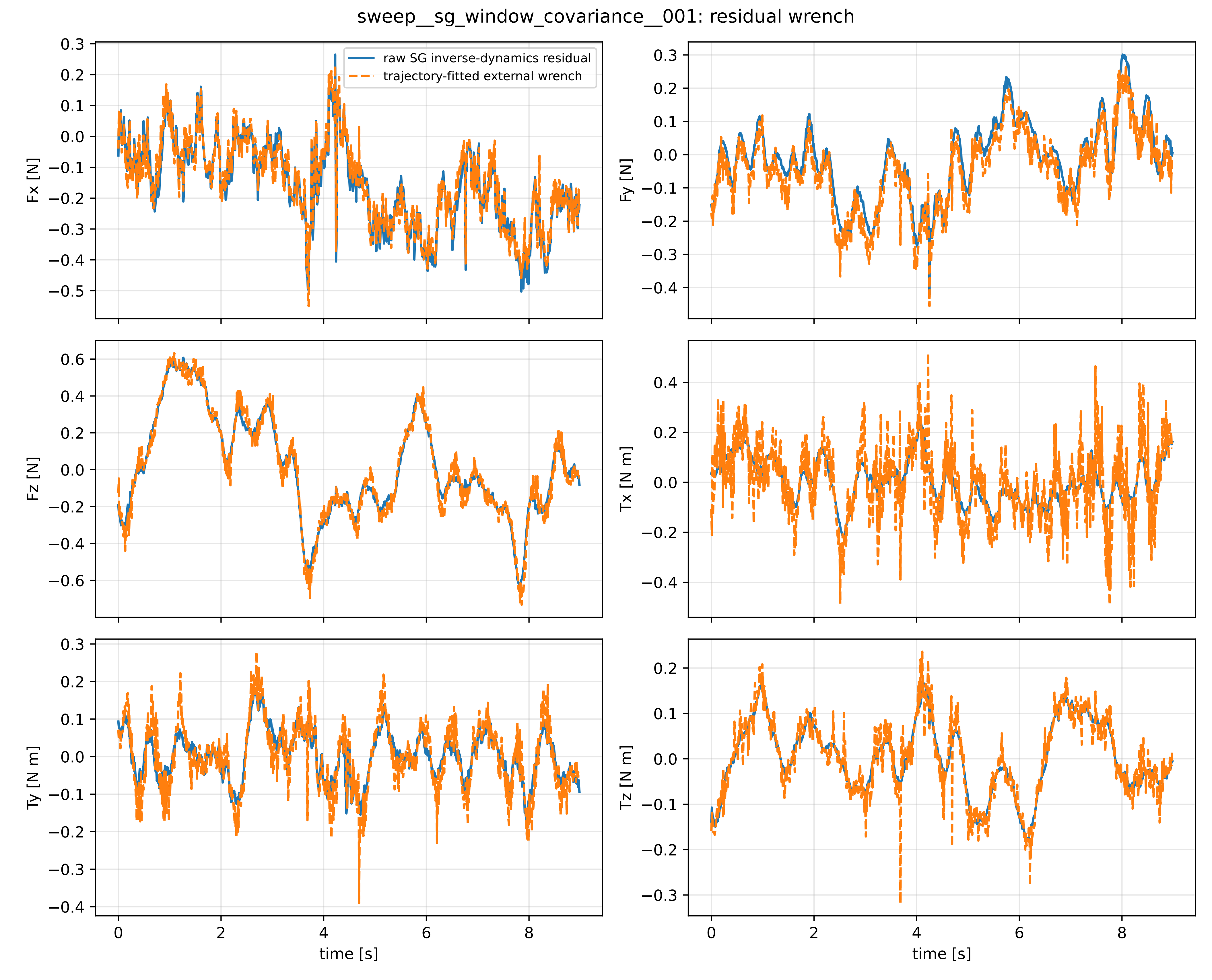
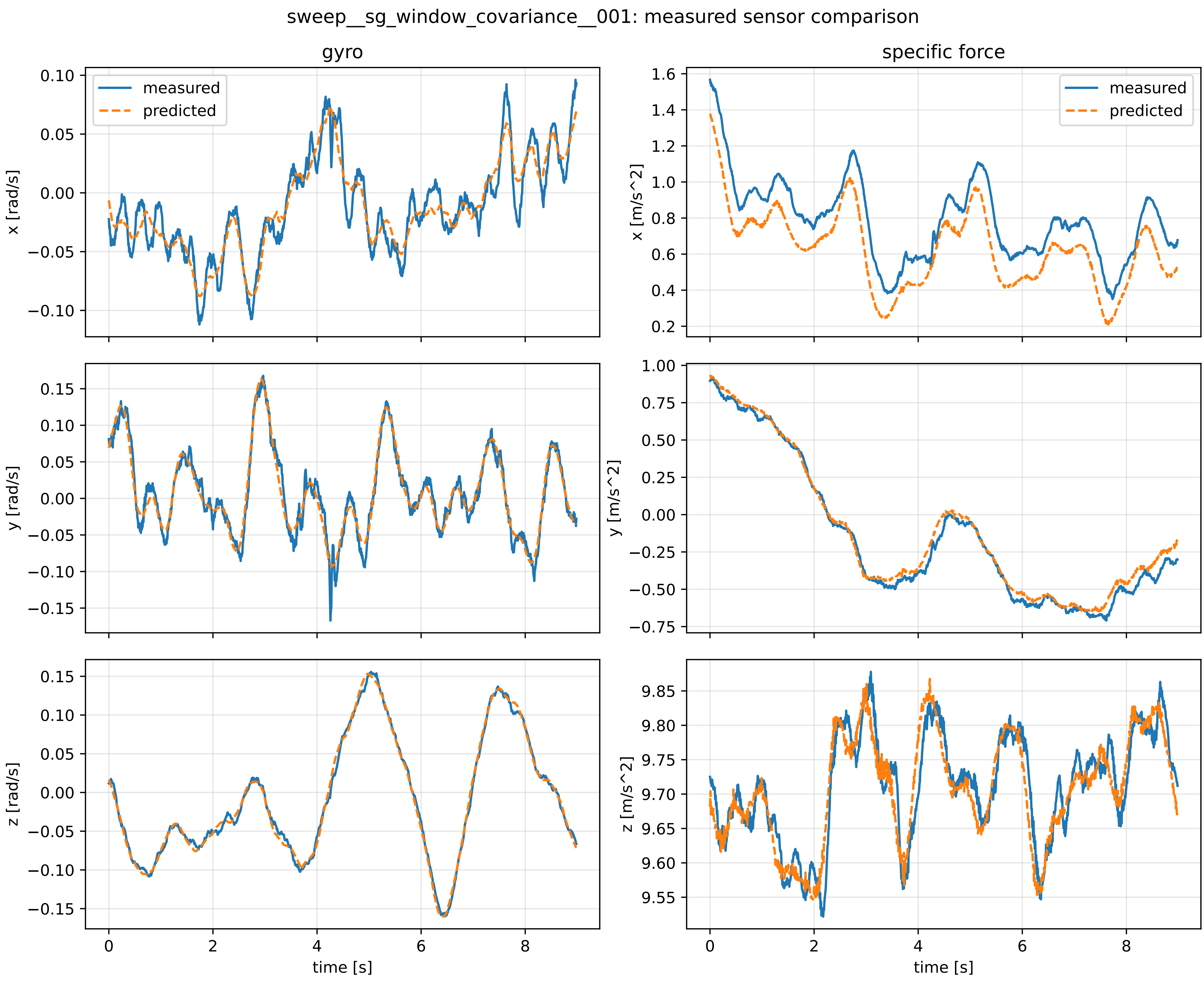
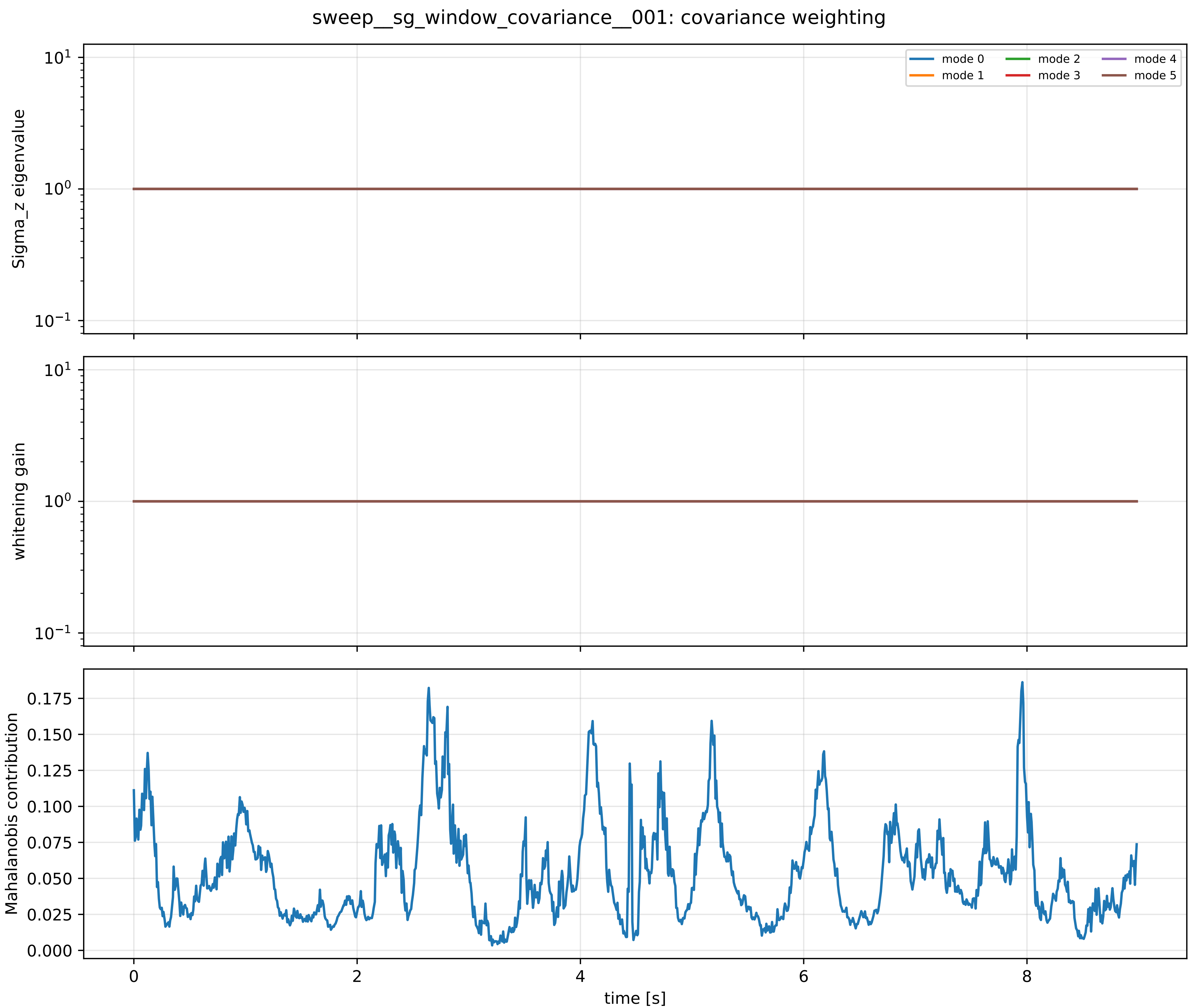

sweep__sg_window_covariance__001: trajectory
observed trajectory
estimated + fitted-wrench reconstruction
x [m]
0.28
0.27
0.26
y [m]
−0.33
−0.34
−0.35
−0.36
−0.37
z [m]
1.08
1.06
1.04
1.02
0
2
4
6
8
time [s]


sweep__sg_window_covariance__001: residual wrench
raw SG inverse-dynamics residual
trajectory-fitted external wrench
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time [s]
time [s]


sweep__sg_window_covariance__001: measured sensor comparison
gyro
specific force
measured
predicted
x [rad/s]
y [rad/s]
z [rad/s]
x [m/s^2]
y [m/s^2]
z [m/s^2]
time [s]
time [s]

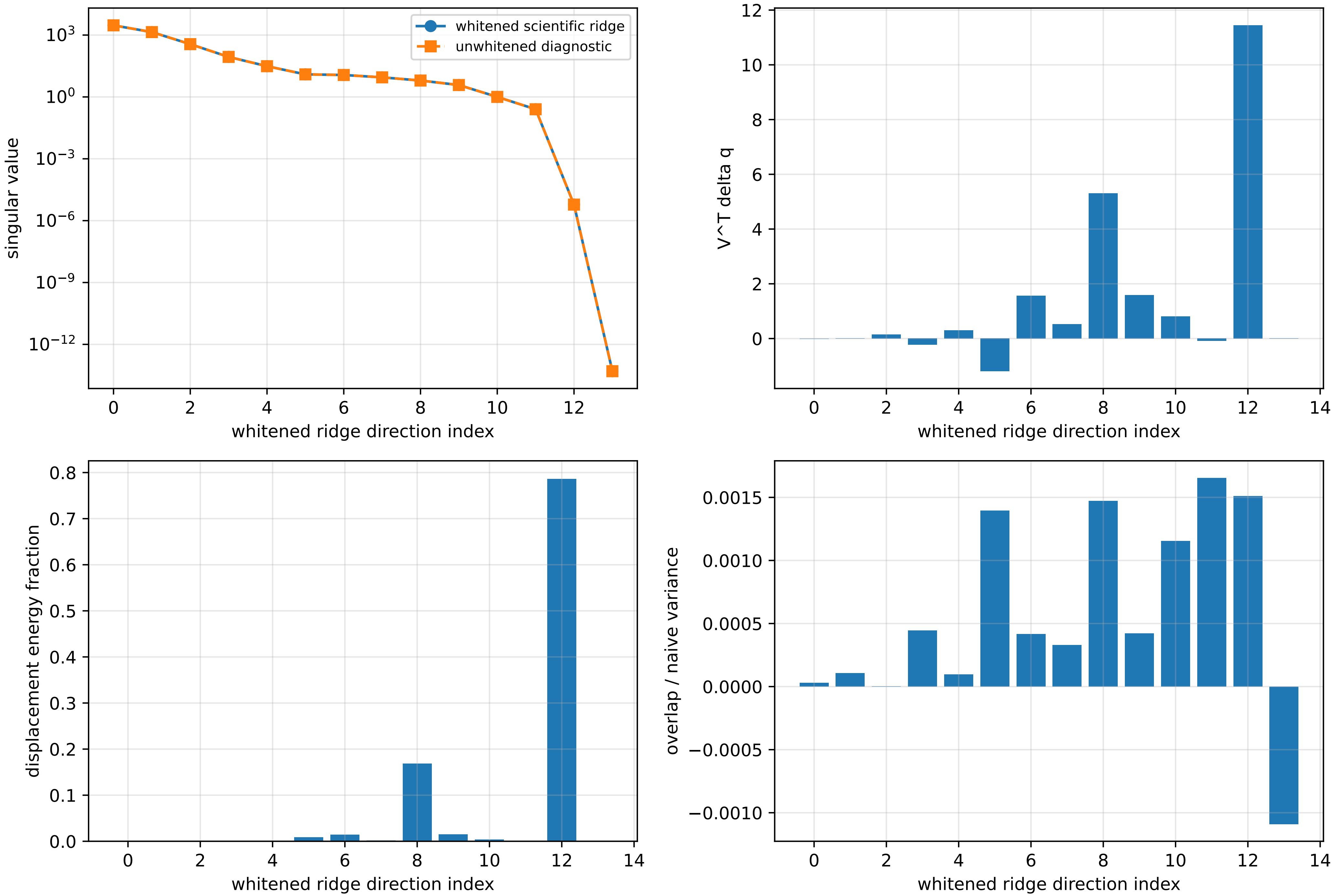
sweep__sg_window_covariance__001: nominal -> estimated parameters

```
Case: sweep__sg_window_covariance__001
status: completed (all stages completed)

mass [kg]       nominal  2.35159759   estimated  4.05977885   delta  1.70818126

inertia [kg m^2] (nominal -> estimated; delta)
  [ 6.5000061e-02 -7.2789925e-07  1.9015080e-05] -> [ 0.9909033 -0.016744   0.0712481]   d=[ 0.9259032 -0.0167433  0.0712291]
  [-7.2789925e-07  6.4952656e-02  5.9167305e-08] -> [-0.016744   0.4454943 -0.0102389]   d=[-0.0167433  0.3805416 -0.0102389]
  [1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [ 0.0712481 -0.0102389  0.5700683]   d=[ 0.0712291 -0.0102389  0.4410762]

CoG [m]       [-2.0247086e-03 -3.0526579e-05  9.5097496e-03] -> [-0.0197021 -0.0580223 -0.0066071]   d=[-0.0176774 -0.057991
rotor force effectiveness
  [1. 1. 1. 1.] -> [1.8117038 1.7969122 0.9049409 1.1219072]   d=[ 0.8117038  0.7969122 -0.0950591  0.1219072]

rotor lag [s]  0.175654623
gimbal lag [s]  0.0400004388
```

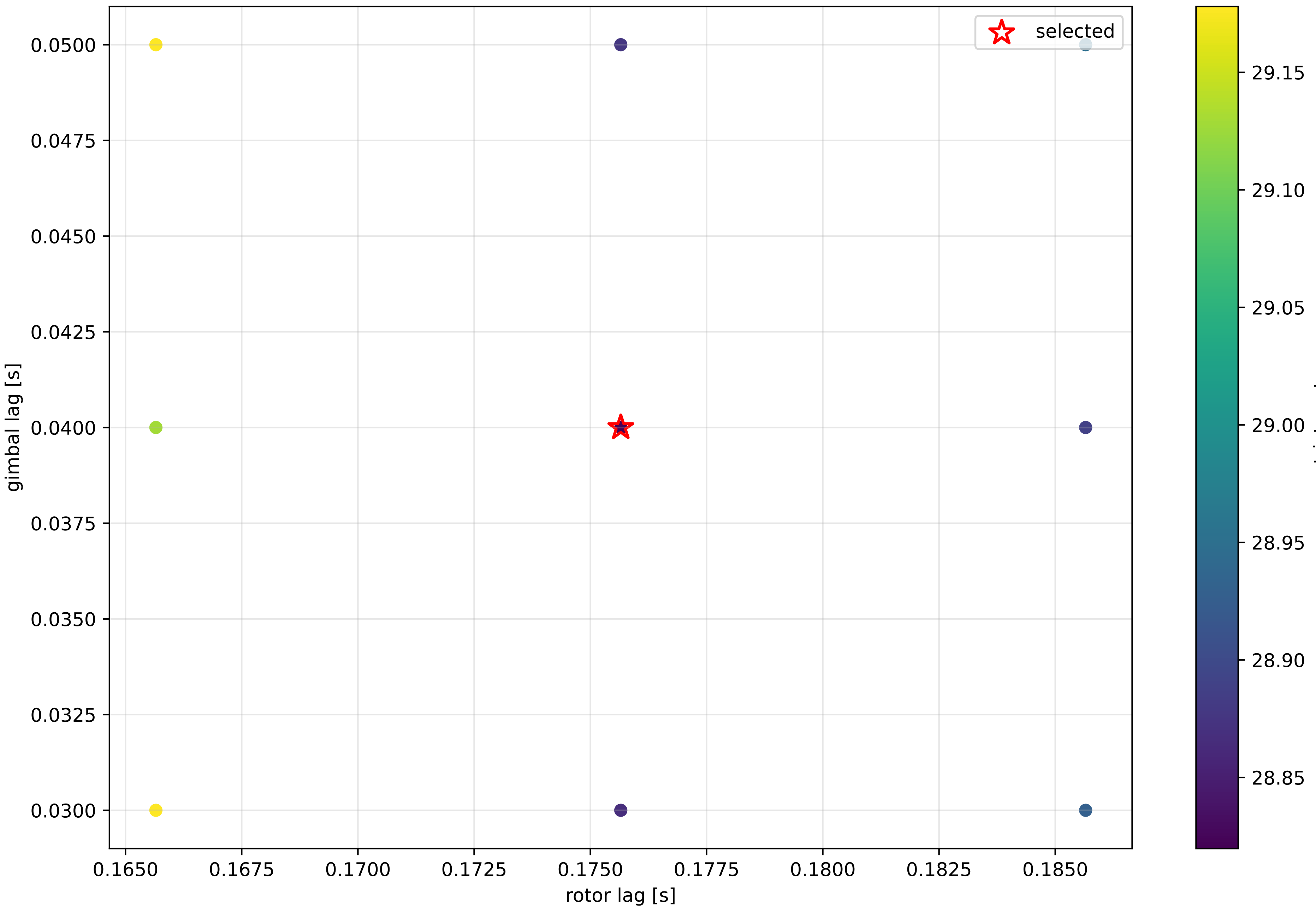


sweep__sg_window_covariance__001: covariance weighting
Sigma_z eigenvalue
whitening gain
Mahalanobis contribution
time [s]
mode 0
mode 1
mode 2
mode 3
mode 4
mode 5
10^1
10^0
10^{-1}
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0
2
4
6
8

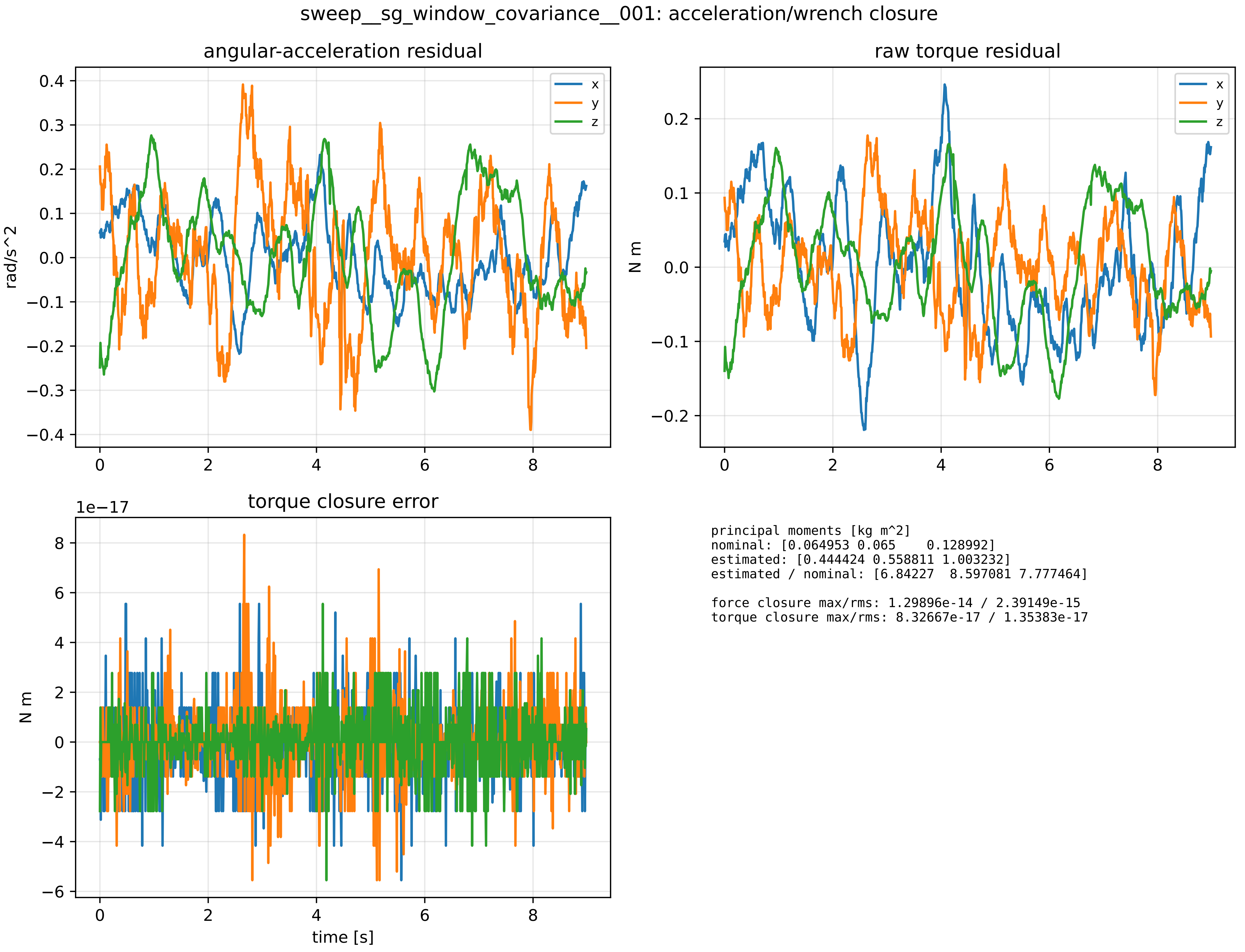
sweep__sg_window_covariance__001: parameter ridge



exact scale gauge: ||J v_scale||=8.22986e-14, rank=13, nullity=1 sweep__sg_window_covariance__001: strict lag candidates



boundary flags: rotor lower=False upper=False; gimbal lower=False upper=False


sweep__sg_window_covariance__001: acceleration/wrench closure
angular-acceleration residual
rad/s^2
raw torque residual
N m
torque closure error
1e−17
N m
time [s]
x
y
z
principal moments [kg m^2]
nominal: [0.064953 0.065   0.128992]
estimated: [0.444424 0.558811 1.003232]
estimated / nominal: [6.84227 8.597081 7.777464]
force closure max/rms: 1.29896e-14 / 2.39149e-15
torque closure max/rms: 8.32667e-17 / 1.35383e-17